

# Strict Global PF<sub>4</sub> for the Riemann Kernel and a Continuous Fourier Separator

Joshua Rainstar

Airspy, Inc.

joshuah.rainstar@gmail.com

16 July 2026

## Abstract

Let  $\Phi$  be the classical Riemann, or de Bruijn–Newman, kernel. We prove that every translation minor of  $\Phi$  of order at most four is strictly positive at strictly ordered nodes, and an exact origin-confluent calculation gives a negative order-five minor. Thus the exact global Polya-frequency order is four. For the positive theorem, a two-mode argument proves global positivity of the log-curvature quantities  $q, F_2$  and of the fully confluent invariant  $\mathcal{C}_4$ . The finite verification uses only rational polynomial algebra, Taylor inequalities, and geometric tails. The order-five origin certificate likewise uses exact rational arithmetic. A quotient-Wronskian argument reduces arbitrary fourth-order minors to  $\partial_\xi \Psi < 0$ . In the increasing curvature coordinate, its numerator is

$$\delta\Lambda \int_p^w W(t) \frac{\mathcal{C}_4(t)}{Q(t)^6 \kappa(t)^2} dt.$$

Here  $W$  is the difference of two explicitly normalized triangular CDFs. Its strict positivity follows from a one-crossing density ratio. The transport reductions and analytic sign estimates are given explicitly, with a table of the finite polynomial inequalities used for the three global signs. We then construct an explicit positive even Schwartz kernel which is itself strictly PF<sub>4</sub> but whose entire Fourier transform has nonreal zeros; the latter conclusion follows from a short exact discriminant inequality. Thus continuous PF<sub>3</sub> and PF<sub>4</sub> membership alone do not force Fourier real-rootedness.

**2020 Mathematics Subject Classification.** 15B48 (primary); 42A38, 30D15, 11M26 (secondary).

**Keywords.** Polya-frequency function, total positivity, Riemann kernel, entire function, characteristic function, Wronskian, Fourier transform, computer-assisted proof.

## Contents

|   |                                               |   |
|---|-----------------------------------------------|---|
| 1 | Introduction                                  | 2 |
| 2 | Relation to earlier work                      | 3 |
| 3 | The kernel and smooth even continuation       | 4 |
| 4 | Certified one-variable inputs                 | 4 |
| 5 | Order three and the positive slope functional | 5 |

|      |                                                          |    |
|------|----------------------------------------------------------|----|
| 6    | Quotient-Wronskian reduction                             | 6  |
| 7    | Curvature coordinate and sign bridge                     | 7  |
| 8    | The confluent invariant in curvature form                | 8  |
| 9    | Exact transport identity                                 | 9  |
| 10   | The positive crossing kernel                             | 9  |
| 11   | Completion and exact order                               | 10 |
| 12   | An explicit continuous $\text{PF}_4$ separator           | 12 |
| 12.1 | Strict $\text{PF}_3$ . . . . .                           | 12 |
| 12.2 | Exact order-four density . . . . .                       | 12 |
| 12.3 | Nonreal Fourier zeros . . . . .                          | 13 |
| 13   | Exact verification and reproduction                      | 13 |
| 14   | Declarations, availability, and computational provenance | 14 |
| A    | Confluent and curvature algebra                          | 15 |
| B    | Exact proof of the certified input                       | 16 |
| B.1  | Cleared sign polynomials . . . . .                       | 16 |
| B.2  | Exact two-mode margin . . . . .                          | 17 |
| B.3  | All later modes . . . . .                                | 18 |
| C    | Endpoint algebra for the transport cancellation          | 19 |
| D    | Certificate inventory                                    | 19 |
| E    | Exact separator calculation                              | 20 |

## 1 Introduction

A function  $f : \mathbb{R} \rightarrow \mathbb{R}$  is  $\text{PF}_r$  when every translation-kernel minor

$$\det[f(x_i - y_j)]_{i,j=1}^k$$

is nonnegative for  $1 \leq k \leq r$  and increasing node tuples. We call the property strict when the determinant is positive whenever both tuples are strictly increasing. This paper determines the exact finite Polya-frequency order of the Riemann kernel; see [3] for modern background and Schoenberg's characterization. Related finite-order and Riemann-kernel work is placed in Section 2.

**Theorem 1.1** (Strict global  $\text{PF}_4$ ). *For every  $1 \leq k \leq 4$  and strictly increasing real tuples  $x_1 < \dots < x_k$  and  $y_1 < \dots < y_k$ ,*

$$\det[\Phi(x_i - y_j)]_{i,j=1}^k > 0.$$

A parity factorization of the fully confluent derivative matrix at the origin gives a negative order-five limit. Divided differences then transfer that sign to equally spaced translation minors at sufficiently small positive spacing; see Theorem 11.2. Hence Theorem 1.1 has the following exact consequence.

**Corollary 1.2.** *The exact global Polya-frequency order of  $\Phi$  is four.*

The exact-order classification and the Fourier question are distinct. At the proved order-four level, the following example shows that finite-order membership alone does not force Fourier real-rootedness, even within the positive even Schwartz class.

**Theorem 1.3** (Continuous  $\text{PF}_4$  Fourier separator). *There is an explicit positive even Schwartz function  $f_*$  which is strict  $\text{PF}_4$ , while its entire Fourier transform has nonreal zeros. Consequently continuous  $\text{PF}_3$  and continuous  $\text{PF}_4$  membership are each insufficient to force Fourier real-rootedness.*

The proof has two one-variable inputs and one exact transport mechanism. First, global inequalities  $q > 0$  and  $F_2 > 0$  imply strict  $\text{PF}_3$  and positivity of a three-point functional  $\Lambda$ . Second, the fully confluent order-four invariant  $\mathcal{C}_4$  is globally positive. Exact quotient identities reduce strict  $\text{PF}_4$  to the monotonicity of a scalar three-point function  $\Psi$ . An increasing curvature coordinate then turns the derivative of  $\Psi$  into a positive crossing-kernel integral of  $\mathcal{C}_4$ .

The finite part of the first input consists of exact rational Bernstein coefficients on fixed algebraic domains, followed by analytic bounds for all remaining theta modes; Appendix B lists the domains, degrees, and minimum coefficients. The transport algebra is proved in the text, and the order-five calculation uses rational enclosures and elementary series bounds. Theorem 1.3, proved in Section 12, supplies an explicit order-four counterexample rather than a finite-order sufficiency threshold.

## 2 Relation to earlier work

Schoenberg introduced Polya-frequency functions and connected translation minors with bilateral Laplace transforms and variation-diminishing convolution operators [1, 3]. Karlin’s monograph remains the standard systematic account of total positivity and its composition principles [2]; a modern review of PF functions and their preservers is given by Belton, Guillot, Khare, and Putinar [3].

Finite-order translation kernels have a substantially wider range than the infinite-order class. Khare’s characterization of measurable  $\text{TN}_p$  functions for  $p \geq 3$  provides a general finite-order setting [4]. The present proof instead derives a criterion specific to order four, using successive quotient derivatives and a curvature transport identity.

The theta normalization used here is the classical one in de Bruijn’s study of the Riemann Fourier kernel [5]. For the same kernel, Dimitrov and Xu relate Wronskians of Fourier and Laplace transforms to correlation kernels and real-zero criteria [6], while Csordas and Varga obtain strict concavity and moment inequalities connected with the Riemann Hypothesis [7]. Grochenig explains the distinct Schoenberg correspondence relevant to the Riemann Hypothesis: the PF function there is recovered from the reciprocal of a Laguerre–Polya entire function, rather than identified with the original Riemann kernel [8].

A preliminary version of the present work containing a  $\text{PF}_5$  counterexample was sent to a publisher in July 2025. The order-five argument below uses an exact origin-confluent sign. The classification proved here is global strict  $\text{PF}_4$ , hence exact order four; Theorem 1.3 separately gives a continuous finite-order Fourier separator.

### 3 The kernel and smooth even continuation

Put

$$\vartheta(x) = \sum_{n \in \mathbb{Z}} e^{-\pi n^2 x}, \quad \Theta(t) = \vartheta(e^{2t}), \quad H(t) = e^{t/2} \Theta(t).$$

The theta transformation  $\vartheta(x) = x^{-1/2} \vartheta(1/x)$  gives

$$\Theta(-t) = e^t \Theta(t), \quad H(-t) = H(t).$$

The series and all its derivatives converge locally uniformly, so  $H$  is real analytic and even. Define

$$\Phi(t) = \frac{1}{2} \left( H''(t) - \frac{1}{4} H(t) \right).$$

Termwise differentiation gives, for every real  $t$ ,

$$\Phi(t) = \sum_{n \geq 1} \left( 4\pi^2 n^4 e^{9t/2} - 6\pi n^2 e^{5t/2} \right) e^{-\pi n^2 e^{2t}}. \quad (3.1)$$

Thus the positive-side formula often written with  $|t|$  is the restriction of a real-analytic even function. In particular, smoothness at the origin is not inferred from numerical coverage.

For  $t \geq 0$ , the coefficient of the  $n$ -th exponential can be factored as

$$2\pi n^2 e^{5t/2} (2\pi n^2 e^{2t} - 3) > 0,$$

because  $2\pi > 3$ . Hence every term in (3.1) is positive and  $\Phi(t) > 0$  on  $[0, \infty)$ . Evenness gives  $\Phi(t) > 0$  on all of  $\mathbb{R}$ , so the following logarithm is globally defined.

Write

$$\ell = \log \Phi, \quad s = \ell', \quad q = -s' = -\ell''.$$

For  $a < b$ , define

$$A(a, b) = s(a) - s(b), \quad M(a, b) = \frac{q(b) - q(a)}{A(a, b)}. \quad (3.2)$$

When  $q > 0$ ,  $s$  is strictly decreasing and  $A(a, b) > 0$ .

### 4 Certified one-variable inputs

Define

$$F_1 = qq'' - (q')^2, \quad F_2 = q^3 - F_1.$$

For the logarithmic cumulants  $c_j = \ell^{(j)}$ , set

$$\begin{aligned} m_0 &= 1, & m_1 &= 0, & m_2 &= c_2, & m_3 &= c_3, \\ m_4 &= c_4 + 3c_2^2, & m_5 &= c_5 + 10c_2c_3, \\ m_6 &= c_6 + 15c_2c_4 + 10c_3^2 + 15c_2^3, \end{aligned}$$

and

$$\mathcal{C}_4(t) = \det[m_{i+j-2}(t)]_{i,j=1}^4. \quad (4.1)$$

**Lemma 4.1** (Certified input). *For the kernel  $\Phi$ ,*

$$q(t) > 0, \quad F_2(t) > 0, \quad \mathcal{C}_4(t) > 0 \quad (t \in \mathbb{R}).$$

The proof combines exact rational polynomial inequalities with analytic theta-tail bounds. For  $u \geq 0$ , put

$$x = \pi e^{2u}, \quad y = e^{-3x}, \quad P_0(z) = 2z - 3,$$

and recursively

$$P_{j+1}(z) = \left(\frac{5}{2} - 2z\right) P_j(z) + 2zP'_j(z).$$

After division by the positive first theta mode, the first two modes give the normalized jet entries

$$a_j(x, y) = \frac{P_j(x) + 4yP_j(4x)}{P_0(x)}.$$

Exact rational Bernstein coefficients on three fixed algebraic domains prove

$$q_{1,2} > 10, \quad (F_2)_{1,2} > 1000, \quad (\mathcal{C}_4)_{1,2} > 50,000,000.$$

The complete contribution of modes  $n \geq 3$  is then bounded by one monotone polynomial-geometric estimate. Its changes in the cleared sign numerators are smaller than the three displayed margins, both on  $\pi \leq x \leq 5$  and on  $x \geq 5$ . Evenness completes the real line. All inequalities, including the derivative envelopes used on the tail, are derived in Appendix B. The exact replay uses no floating point, no interval subdivision, and no FLINT or Arb arithmetic.

## 5 Order three and the positive slope functional

For  $\xi < m < r$ , define

$$\Lambda(\xi; m, r) = A(\xi, r) + M(\xi, m) - M(m, r). \quad (5.1)$$

We give the full estimate used to prove its positivity.

Set  $f_0 = q'/q$ . Because  $s' = -q$ ,

$$M(a, b) = \frac{\int_a^b q'(t) dt}{\int_a^b q(t) dt} = \frac{\int_a^b f_0(t) q(t) dt}{\int_a^b q(t) dt};$$

thus  $M(a, b)$  is a  $q$ -weighted mean of  $f_0$ . Choose a point  $u \in [\xi, m]$  where  $f_0$  is minimal on the left interval and a point  $v \in [m, r]$  where it is maximal on the right. Then

$$M(m, r) - M(\xi, m) \leq f_0(v) - f_0(u) \leq \int_{\xi}^r \max(f'_0(t), 0) dt.$$

Since

$$f'_0 = \frac{qq'' - (q')^2}{q^2} = \frac{F_1}{q^2}$$

and  $A(\xi, r) = \int_{\xi}^r q(t) dt$ , equation (5.1) yields

$$\Lambda(\xi; m, r) \geq \int_{\xi}^r \left( q - \frac{\max(F_1, 0)}{q^2} \right) dt \quad (5.2)$$

$$= \int_{\xi}^r \frac{q^3 - \max(F_1, 0)}{q^2} dt = \int_{\xi}^r \frac{\min(q^3, F_2)}{q^2} dt > 0. \quad (5.3)$$

The last inequality follows from Lemma 4.1 and  $q > 0$ .

The same quantity is the logarithmic slope derivative in the normalized order-three determinant. Hence (5.3), together with strict log-concavity, proves strict global PF<sub>3</sub>. We rederive the needed Wronskian form in the next section because its signs also control order four.

## 6 Quotient-Wronskian reduction

Fix  $y_1 < y_2 < y_3 < y_4$ , put  $u_j(t) = \Phi(t - y_j)$ , and write  $p_j = t - y_j$ , so  $p_4 < p_3 < p_2 < p_1$ . Define

$$v_j = (u_j/u_1)', \quad w_j = (v_j/v_2)'.$$

Successive column division and differentiation give

$$W(u_1, u_2, u_3) = u_1^3 W(v_2, v_3), \quad (6.1)$$

$$W(u_1, u_2, u_3, u_4) = u_1^4 v_2^3 w_3^2 \frac{d}{dt} \left( \frac{w_4}{w_3} \right). \quad (6.2)$$

Let  $\mathcal{T}$  denote simultaneous translation of all point arguments. From (3.2),

$$\mathcal{T}A(a, b) = q(b) - q(a), \quad \mathcal{T} \log A(a, b) = M(a, b). \quad (6.3)$$

The sign convention in (6.3) will be used below. Since

$$\frac{v_j}{v_2} = \frac{u_j}{u_2} \frac{A(p_j, p_1)}{A(p_2, p_1)},$$

direct differentiation gives

$$g_j := \mathcal{T} \log(v_j/v_2) = A(p_j, p_2) + M(p_j, p_1) - M(p_2, p_1). \quad (6.4)$$

The exact weighted-mean split

$$A(p_j, p_1)M(p_j, p_1) = A(p_j, p_2)M(p_j, p_2) + A(p_2, p_1)M(p_2, p_1)$$

turns (6.4) into

$$g_j = \frac{A(p_j, p_2)}{A(p_j, p_1)} \Lambda(p_j; p_2, p_1) > 0. \quad (6.5)$$

Thus  $w_j = (v_j/v_2)g_j > 0$ , and (6.1) has the strict order-three sign.

Define

$$\Psi(\xi; m, r) = \Lambda(\xi; m, r) + \mathcal{T} \log \Lambda(\xi; m, r).$$

Because  $w_j = (v_j/v_2)g_j$ ,

$$\mathcal{T} \log(w_4/w_3) = (g_4 + \mathcal{T} \log g_4) - (g_3 + \mathcal{T} \log g_3). \quad (6.6)$$

Equation (6.5) and (6.3) give

$$\mathcal{T} \log g_j = M(p_j, p_2) - M(p_j, p_1) + \mathcal{T} \log \Lambda_j,$$

where  $\Lambda_j = \Lambda(p_j; p_2, p_1)$ . A second use of the same mean split gives

$$g_j + M(p_j, p_2) - M(p_j, p_1) = \Lambda_j - A(p_2, p_1).$$

The common last term cancels in (6.6), yielding the exact identity

$$\mathcal{T} \log(w_4/w_3) = \Psi(p_4; p_2, p_1) - \Psi(p_3; p_2, p_1). \quad (6.7)$$

**Lemma 6.1** (Iterated quotient integral). *Let  $f_1, \dots, f_k$  be continuously differentiable on an interval, let  $f_1 > 0$ , and put  $G_j = f_j/f_1$ . For  $t_1 < \dots < t_k$ ,*

$$\det[f_j(t_i)]_{i,j=1}^k = \prod_{i=1}^k f_1(t_i) \int_{t_1}^{t_2} \cdots \int_{t_{k-1}}^{t_k} \det[G'_{j+1}(\tau_i)]_{i,j=1}^{k-1} d\tau_1 \cdots d\tau_{k-1}. \quad (6.8)$$

*The identity may be applied repeatedly whenever the first member of the new derivative family is positive.*

*Proof.* Factor  $f_1(t_i)$  from row  $i$ . The first column is then one. Replace successive rows, from bottom to top, by their difference with the preceding row. Each remaining entry is  $G_j(t_{i+1}) - G_j(t_i) = \int_{t_i}^{t_{i+1}} G'_j(\tau) d\tau$ . Determinant multilinearity gives (6.8). Since  $\tau_i \in [t_i, t_{i+1}]$ , the integration nodes remain ordered, so the same argument applies to the derivative determinant.  $\square$

**Proposition 6.2** (Three-point equivalence). *Under the strict lower-order positivity already proved, global  $\text{PF}_4$  is equivalent to*

$$\partial_\xi \Psi(\xi; m, r) \leq 0 \quad (\xi < m < r).$$

*Strict inequality implies strict global  $\text{PF}_4$ .*

*Proof.* If  $\partial_\xi \Psi \leq 0$ , then  $p_4 < p_3$  makes the right side of (6.7) nonnegative. All prefactors in (6.2) are positive. The Wronskian is therefore nonnegative, and is positive in the strict case.

For clarity, applying Lemma 6.1 three times to the order-four minor gives, with  $I_i = [t_i, t_{i+1}]$ ,

$$\begin{aligned} \det[u_j(t_i)]_{i,j=1}^4 &= \prod_{i=1}^4 u_1(t_i) \int_{I_1 \times I_2 \times I_3} \prod_{i=1}^3 v_2(\tau_i) \\ &\quad \times \int_{\tau_1}^{\tau_2} \int_{\tau_2}^{\tau_3} w_3(\sigma_1) w_3(\sigma_2) \int_{\sigma_1}^{\sigma_2} \left( \frac{w_4}{w_3} \right)'(\rho) d\rho d\sigma_2 d\sigma_1 d\tau. \end{aligned}$$

Every factor is positive in the strict case, and every integration interval has positive length. This proves positivity of every ordered collocation minor, not only of the confluent Wronskian.

Conversely, assume global  $\text{PF}_4$ . Let all four row nodes coalesce, divide by their positive Vandermonde, and pass to the limit. The resulting translate Wronskian is nonnegative. Equations (6.2) and (6.7) imply  $\Psi(p_4; p_2, p_1) \geq \Psi(p_3; p_2, p_1)$  whenever  $p_4 < p_3$ . Thus  $\Psi(\cdot; m, r)$  is nonincreasing, which is equivalent to the stated differential inequality because it is smooth.  $\square$

## 7 Curvature coordinate and sign bridge

Since  $q > 0$ ,

$$y(t) = -s(t), \quad Q(y(t)) = q(t)$$

defines a global strictly increasing coordinate on its image, with  $dy/dt = q(t) = Q(y) > 0$ . Primes in this and the next three sections mean  $y$ -derivatives. Put

$$\rho = 1 - Q'' = \frac{F_2}{q^3} > 0, \quad \kappa = 2 - Q'' = 1 + \rho > 1.$$

For  $\xi < m < r$ , write

$$p = y(\xi), \quad z = y(m), \quad w = y(r), \quad L = z - p, \quad R = w - z.$$

Because  $A(a, b) = y(b) - y(a)$  and  $M(a, b) = Q[y(a), y(b)]$ ,

$$\Lambda = (w - p) + Q[p, z] - Q[z, w]. \quad (7.1)$$

Two integrations of  $\kappa = 2 - Q''$  give

$$\Lambda = \frac{1}{L} \int_p^z (y - p) \kappa(y) dy + \frac{1}{R} \int_z^w (w - y) \kappa(y) dy, \quad (7.2)$$

$$\delta := -\partial_p \Lambda = \frac{1}{L^2} \int_p^z (z - y) \kappa(y) dy > 0. \quad (7.3)$$

Simultaneous translation becomes

$$\mathcal{T} = Q(p) \partial_p + Q(z) \partial_z + Q(w) \partial_w,$$

and  $\partial_\xi = Q(p) \partial_p$ . Thus  $\Lambda_\xi = -Q(p) \delta$ .

Define

$$\mathcal{N} = \delta \Lambda^2 + Q'(p) \delta \Lambda + \Lambda \mathcal{T} \delta - \delta \mathcal{T} \Lambda. \quad (7.4)$$

Differentiating  $\Psi = \Lambda + \mathcal{T} \log \Lambda$ , commuting simultaneous translation with  $\partial_\xi$ , and collecting over  $\Lambda^2$  gives

$$\partial_\xi \Psi = -\frac{Q(p)}{\Lambda^2} \mathcal{N}. \quad (7.5)$$

Equivalently,

$$\frac{\mathcal{N}}{\delta \Lambda} = K := \Lambda + Q'(p) + \mathcal{T} \log \delta - \mathcal{T} \log \Lambda. \quad (7.6)$$

It remains to prove  $K > 0$ .

## 8 The confluent invariant in curvature form

Define

$$D = 3(\kappa - 1) - \{Q(\log \kappa)'\}' \quad (8.1)$$

**Lemma 8.1** (Confluent-to-curvature identity). *The determinant (4.1) satisfies*

$$\mathcal{C}_4 = Q^6 \kappa^2 D. \quad (8.2)$$

Consequently  $D > 0$  on  $\mathbb{R}$ .

*Proof.* The change of variable gives  $d/dt = Q d/dy$  and  $c_2 = -Q$ . Recursively applying this operator produces the five cumulants listed in Appendix A. Substitution into the Schur-complement form of (4.1) factors out  $Q^6$ . The remaining factor is

$$\kappa^2 (3(\kappa - 1) - \{Q(\log \kappa)'\}'),$$

as shown by the common expanded numerator in Appendix A. This proves (8.2) without a numerical assumption. Lemma 4.1,  $Q > 0$ , and  $\kappa > 0$  then imply  $D > 0$ .  $\square$

The determinant definition is primary: it is the doubly confluent fourth-order translation minor after division by the positive row and column Vandermonde factors and by  $\Phi^4$ . The explicit thirteen-term expansion is included in Appendix A for independent checking of the certificate input.



## 9 Exact transport identity

Normalize the triangular weights in (7.2) and (7.3) as probability measures:

$$d\mu(y) = \frac{(z-y)\kappa(y)}{L^2\delta} \mathbf{1}_{[p,z]}(y) dy, \quad (9.1)$$

$$d\nu(y) = \frac{(y-p)\kappa(y)}{L\Lambda} \mathbf{1}_{[p,z]}(y) dy + \frac{(w-y)\kappa(y)}{R\Lambda} \mathbf{1}_{[z,w]}(y) dy. \quad (9.2)$$

Set

$$A_0(y) = 3(y - Q'(y)) - Q(y) \frac{\kappa'(y)}{\kappa(y)}.$$

Then  $A'_0 = D$ .

**Lemma 9.1** (Transport cancellation). *With  $K$  as in (7.6),*

$$K = \mathbb{E}_\nu[A_0] - \mathbb{E}_\mu[A_0]. \quad (9.3)$$

*Proof.* Two elementary primitives carry the complete cancellation:

$$\kappa A_0 = H', \quad H = 3y^2 - 3Q - 3yQ' + (Q')^2 + QQ'', \quad (9.4)$$

$$H = J', \quad J = y^3 - 3yQ + QQ'. \quad (9.5)$$

Both identities follow by one differentiation.

Let  $I_L = J(z) - J(p)$  and  $I_R = J(w) - J(z)$ . Integrating the linear weights in (9.1)–(9.2) by parts once gives

$$\mathbb{E}_\nu[A_0] - \mathbb{E}_\mu[A_0] = \frac{-I_L/L + I_R/R}{\Lambda} + \frac{LH(p) - I_L}{L^2\delta}. \quad (9.6)$$

On the other hand, differentiation of the triangular integrals along  $\mathcal{T}$  gives

$$\mathcal{T}\Lambda = U(z, w) - U(p, z), \quad \mathcal{T}\delta = \frac{U(p, z) - (Q(z) - Q(p))\delta - Q(p)\kappa(p)}{L},$$

where

$$U(c, d) = Q(c) + Q(d) - \frac{Q(d)Q'(d) - Q(c)Q'(c)}{d - c} + \frac{(Q(d) - Q(c))^2}{(d - c)^2}. \quad (9.7)$$

Substitution in (7.6) yields

$$K = \Lambda + Q'(p) - Q[p, z] + \frac{U(p, z) - U(z, w)}{\Lambda} + \frac{U(p, z) - Q(p)\kappa(p)}{L\delta}. \quad (9.8)$$

The endpoint identity equating (9.6) and (9.8), with every substitution exposed, is Appendix C. It is an identity in the endpoint positions and the values of  $Q, Q', Q''$ ; no bound or polynomial model is used. This proves (9.3).  $\square$

## 10 The positive crossing kernel

Let  $F_\mu, F_\nu$  be the CDFs of (9.1)–(9.2), and put

$$W(y) = F_\mu(y) - F_\nu(y).$$

**Lemma 10.1** (Strict stochastic ordering). *One has  $W(y) > 0$  for  $p < y < w$ , while  $W(p) = W(w) = 0$ .*

*Proof.* On  $[z, w]$ ,  $F_\mu = 1$ , so  $W = 1 - F_\nu > 0$  before  $w$ . On  $(p, z)$ , the ratio of the two positive densities is

$$R_0(y) = \frac{d\mu/dy}{d\nu/dy} = \frac{\Lambda(z-y)}{L\delta(y-p)}. \quad (10.1)$$

Writing  $C = \Lambda/(L\delta) > 0$ , direct differentiation gives

$$R'_0(y) = -\frac{C(z-p)}{(y-p)^2} < 0. \quad (10.2)$$

Thus  $R_0$  decreases strictly from  $+\infty$  to 0. Hence  $W' = \mu' - \nu'$  changes sign exactly once, from positive to negative. Since  $W(p) = 0$  and

$$W(z) = 1 - F_\nu(z) = \nu((z, w]) > 0,$$

the function cannot return to zero on  $(p, z]$ . The right interval was already handled.  $\square$

Integration by parts for CDFs, using  $A'_0 = D$  and the common total mass, turns Lemma 9.1 into

$$K = \int_p^w W(y)D(y)dy. \quad (10.3)$$

Combining (7.6), (8.2), and (10.3) gives the central identity

$$\mathcal{N} = \delta\Lambda \int_p^w W(y) \frac{\mathcal{C}_4(y)}{Q(y)^6 \kappa(y)^2} dy > 0. \quad (10.4)$$

Every factor in the interior is strictly positive.

## 11 Completion and exact order

Equation (7.5) and (10.4) imply

$$\partial_\xi \Psi(\xi; m, r) < 0 \quad (\xi < m < r).$$

Proposition 6.2 therefore proves Theorem 1.1.

Michałowski also studied order-five Toeplitz obstructions numerically [9]; here a direct origin-confluent determinant suffices.

We now prove directly that the classification stops at order four. Put

$$H_k = \det[\Phi^{(i+j)}(0)]_{i,j=0}^{k-1}, \quad s_j = \Phi^{(j)}(0).$$

Since  $\Phi$  is even, simultaneous parity permutations of the rows and columns give

$$H_4 = AB, \quad H_5 = CB,$$

where

$$A = s_0 s_4 - s_2^2, \quad B = s_2 s_6 - s_4^2,$$

and

$$C = \det \begin{pmatrix} s_0 & s_2 & s_4 \\ s_2 & s_4 & s_6 \\ s_4 & s_6 & s_8 \end{pmatrix}.$$

**Lemma 11.1.** *For the Riemann kernel,*

$$445 < A < 446, \quad 189223 < B < 189228, \quad -674000000 < C < -614000000.$$

*In particular,  $H_4 > 0$  and  $H_5 < 0$ .*

*Proof.* For  $x = \pi n^2 e^{2u}$ , the  $n$ -th summand of  $\Phi(u)$  is  $e^{u/2-x} P_0(x)$ , where  $P_0(x) = 2x(2x-3)$ . If its  $j$ -th derivative is  $e^{u/2-x} P_j(x)$ , then

$$P_{j+1} = 2xP'_j + (1/2 - 2x)P_j.$$

Thus

$$s_j = \sum_{n \geq 1} e^{-\pi n^2} P_j(\pi n^2).$$

The first three modes are enclosed by rational alternating-series bounds. Machin's identity  $\pi = 16 \arctan(1/5) - 4 \arctan(1/239)$  supplies rational bounds for  $\pi$ , and argument division followed by the degree-29 and degree-30 alternating Taylor sums supplies rational bounds for each exponential.

For the omitted modes, let  $S_j$  be the sum of the absolute coefficients of  $P_j$ . Since  $\deg P_j = j+2$ ,  $3 < \pi < 22/7$ , and  $e^3 > \sum_{r=0}^8 3^r/r! > 20$ ,

$$\sum_{n \geq 4} |P_j(\pi n^2)| e^{-\pi n^2} \leq \frac{S_j (22/7)^{j+2} 4^{2j+4} 20^{-16}}{1 - (5/4)^{20} 20^{-9}} \quad (j = 0, 2, 4, 6, 8).$$

All endpoints lie on  $10^{-36}\mathbb{Z}$ , and every arithmetic operation is rounded outward. Substitution into the displayed formulas gives the entirely rational sign chain

$$\begin{aligned} 0 < 445 < A < 446, \quad 0 < 189223 < B < 189228, \\ -674000000 < C < -614000000 < 0. \end{aligned}$$

Thus  $AB > 0$  and  $CB < 0$  without a floating-point determinant. The full grid endpoints and the standard-library verifier are included in the reproducibility archive.  $\square$

**Theorem 11.2.** *The Riemann kernel is not  $\text{PF}_5$ .*

*Proof.* For a  $C^{2k-2}$  function  $f$ , repeated row and column divided differences give, for increasing fixed tuples  $a, b$ ,

$$\lim_{h \downarrow 0} \frac{\det[f(z + h(a_i - b_j))]_{i,j=0}^{k-1}}{h^{k(k-1)} V(a) V(b)} = \frac{(-1)^{k(k-1)/2}}{\prod_{r=0}^{k-1} (r!)^2} \det[f^{(i+j)}(z)]_{i,j=0}^{k-1}.$$

Take  $k = 5$ ,  $z = 0$ , and  $a_i = b_i = i$ . The sign factor is one, both Vandermonde factors are positive, and Lemma 11.1 makes the limit strictly negative. Hence  $\det[\Phi((i-j)h)]_{i,j=0}^4 < 0$  for every sufficiently small positive  $h$ . These are translation minors at strictly increasing nodes.  $\square$

Theorem 1.1 and Theorem 11.2 prove Corollary 1.2.

*Remark 11.3* (Riemann Hypothesis boundary). Schoenberg's transform theorem says that the bilateral Laplace transform of a  $\text{PF}_\infty$  function is the reciprocal of a Laguerre–Polya entire function on its strip of convergence [1, 3]. The Fourier transform of  $\Phi$ , up to the standard normalization, is the Riemann  $\Xi$ -function, which has known real zeros; hence  $\Phi$  is unconditionally not  $\text{PF}_\infty$ . The Schoenberg formulation related to the Riemann Hypothesis instead asks for the inverse transform of the reciprocal of  $\Xi$  to be a Polya-frequency function [8]. Thus the finite-order classification proved here is a different statement. The next section further shows that strict  $\text{PF}_4$  alone does not imply Fourier real-rootedness.

## 12 An explicit continuous PF<sub>4</sub> separator

We now prove Theorem 1.3. Put

$$c = \frac{1}{64}, \quad B(w) = \sum_{k=-3}^3 b_k w^k, \quad (b_{-3}, \dots, b_3) = (2, 10, 23, 30, 23, 10, 2),$$

and define

$$f_*(x) = e^{-cx^2} B(e^{2cx}) = \sum_{k=-3}^3 b_k e^{ck^2} e^{-c(x-k)^2}. \quad (12.1)$$

The Gaussian-mixture expression proves positivity and Schwartz decay; palindromy of  $(b_k)$  proves evenness.

### 12.1 Strict PF<sub>3</sub>

For fixed  $x$ , give  $K \in \{-3, \dots, 3\}$  probability proportional to  $b_k e^{2ckx}$ , and write  $\kappa_j$  for its cumulants. Differentiation of the finite log-partition function gives

$$q = 2c - (2c)^2 \kappa_2, \quad q' = -(2c)^3 \kappa_3, \quad q'' = -(2c)^4 \kappa_4. \quad (12.2)$$

If  $\mu = \mathbb{E}K$ , then  $0 \leq \mathbb{E}[(K+3)(3-K)] = 9 - \kappa_2 - \mu^2$ . Thus  $0 \leq \kappa_2 \leq 9$ , and hence

$$q \geq 2c(1 - 18c) = 2c \frac{23}{32} > 0. \quad (12.3)$$

Moreover,

$$\begin{aligned} F_2 &= q^3 - \{qq'' - (q')^2\} \\ &= q^3 + q(2c)^4 \kappa_4 + (2c)^6 \kappa_3^2. \end{aligned} \quad (12.4)$$

The identity  $\kappa_4 = \mathbb{E}(K - \mathbb{E}K)^4 - 3\kappa_2^2$  gives  $\kappa_4 \geq -243$ . Dropping the final nonnegative term in (12.4) gives

$$F_2 \geq q\{q^2 - 3888c^4\}.$$

At  $c = 1/64$ , the two coefficients after division by  $c^2$  are

$$q^2/c^2 \geq \frac{529}{256}, \quad 3888c^2 = \frac{243}{256}.$$

Thus  $F_2 \geq qc^2(143/128) > 0$  globally. The weighted-mean criterion of Section 5 proves strict PF<sub>3</sub>.

### 12.2 Exact order-four density

Let  $\epsilon = 2c = 1/32$ ,  $t = e^{\epsilon x}$ , and

$$P(t) = t^3 B(t) = 2 + 10t + 23t^2 + 30t^3 + 23t^4 + 10t^5 + 2t^6.$$

For  $E = t \frac{d}{dt}$ , define

$$N_1 = tP', \quad N_{j+1} = t(N_j'P - jN_jP'). \quad (12.5)$$

The quotient rule gives  $E^j \log P = N_j/P^j$ . Therefore, for  $\ell_j = (\log f_*)^{(j)}$ ,

$$\ell_2 = \frac{-\epsilon P^2 + \epsilon^2 N_2}{P^2}, \quad \ell_j = \frac{\epsilon^j N_j}{P^j} \quad (3 \leq j \leq 6). \quad (12.6)$$

Insert these jets into the central-moment determinant (4.1). Every monomial has derivative weight twelve, so every term may first be put over the common denominator  $P^{12}$ . The resulting numerator has the exact factor  $P^8$ . After cancellation,

$$\mathcal{C}_4[f_*](x) = \frac{\tilde{H}(t)}{2^{54}P(t)^4}. \quad (12.7)$$

The exact expansion printed in Appendix E proves that  $\tilde{H}$  is palindromic of degree 24 and that all 25 integer coefficients are strictly positive. Hence  $\mathcal{C}_4[f_*] > 0$  on  $\mathbb{R}$ .

The established inequalities  $q > 0$  and  $F_2 > 0$  also give

$$\kappa = 2 - Q'' = \frac{q^3 + F_2}{q^3} > 0.$$

The generic transport identity (10.4) now has strictly positive density and crossing weight for  $f_*$ . Thus  $\partial_\xi \Psi < 0$ , and Proposition 6.2 proves that  $f_*$  is strict  $\text{PF}_4$ .

### 12.3 Nonreal Fourier zeros

With  $\hat{f}(z) = \int_{\mathbb{R}} f(x)e^{-izx} dx$ , Gaussian translation gives

$$\hat{f}_*(z) = \sqrt{\frac{\pi}{c}} e^{-z^2/(4c)} \sum_{k=-3}^3 b_k e^{ck^2} e^{-ikz}. \quad (12.8)$$

Put  $w = e^{-iz}$  and  $u = w + w^{-1}$ . The Laurent factor in (12.8) vanishes exactly when

$$R_c(u) = 2e^{9c}u^3 + 10e^{4c}u^2 + (23e^c - 6e^{9c})u + 30 - 20e^{4c} \quad (12.9)$$

vanishes. At  $c = 1/64$ , the exact calculation in Appendix E gives  $\text{disc}(R_c) < 0$ . Thus the real cubic has a nonreal conjugate pair. For  $|w| = 1$ , however,  $w + w^{-1}$  is real. The corresponding  $w$ -roots therefore lie off the unit circle and lift through  $w = e^{-iz}$  to zeros with  $\text{Im } z \neq 0$ . The Gaussian factor is zero-free. This completes the proof of Theorem 1.3.

## 13 Exact verification and reproduction

The repository supplies five independent verifiers for the finite algebra and rational inequalities used above.

| Object                         | Repository verifier                                     | Arithmetic          |
|--------------------------------|---------------------------------------------------------|---------------------|
| Global $q, F_2, \mathcal{C}_4$ | <code>scripts/verify_riemann_signs_exact.py</code>      | rational polynomial |
| Quotient reduction             | <code>scripts/verify_pf4_wronskian_reduction.py</code>  | symbolic exact      |
| Transport identity             | <code>scripts/verify_pf4_transport_kernel.py</code>     | symbolic exact      |
| Origin order-five sign         | <code>scripts/verify_pf5_origin.py</code>               | rational enclosures |
| Continuous separator           | <code>scripts/verify_continuous_pf4_separator.py</code> | rational polynomial |

The global-sign verifier retains the first two theta modes exactly. It checks the 361 Bernstein and decay inequalities summarized in Appendix B, the 140 derivative-envelope coefficients, and the analytic geometric bounds for all modes  $n \geq 3$ . Its launcher selects SymPy's pure-Python rational domain and refuses a FLINT import. No floating-point comparison or interval subdivision decides a sign.

The order-five verifier is independent of that calculation. It uses integer arithmetic with outward rounding on  $10^{-36}\mathbb{Z}$ , Machin bounds for  $\pi$ , alternating Taylor bounds for three theta modes, and the geometric tail in Lemma 11.1. The separator verifier reconstructs the central determinant, checks the exact  $P^8$  cancellation, verifies all 25 coefficients in (12.7), and reproduces the rational discriminant inequality in Appendix E. The generated coefficient record has SHA-256

`d0f91259919c7f237ef65068041122656636ca990fe6ef41a146dd53eaa1dba9.`

The quotient and transport scripts check identities for generic smooth jets. Their numerical evaluations at selected points are regression tests only; all sign conclusions follow from the displayed identities and exact inequalities.

From a clean checkout with Python 3.13, SymPy 1.14.0, and mpmath 1.3.0, the single replay command is

`python scripts/replay_paper.py.`

It prints five lines beginning PASS, one for each row above, followed by `status=paper evidence replay passed`. The SHA-256 of its complete standard output is

`ec48df7f12724a948075be71d1b6771e8fb2ea735ab74fb490ab61eafcb7eea3.`

Package versions and the complete artifact hashes are also recorded in `requirements-paper.txt` and `paper/RELEASE_MANIFEST.md`.

## 14 Declarations, availability, and computational provenance

**Author contribution.** Joshua Rainstar conceived and directed the project, developed the proof architecture, verified the exact computations, and prepared the manuscript.

**Acknowledgments and computational assistance.** The author acknowledges the use of OpenAI Codex and other disclosed AI systems for symbolic calculation, implementation, drafting, and review. The author takes responsibility for the mathematical claims and the final manuscript.

**Conflict of interest.** The author declares no competing interests.

**Code and data availability.** The public repository is <https://github.com/falseywinchnet/zeta>. The exact analytic proof was promoted in commit `0ebb95b141a1`. The submission source, generated coefficient table, environment lock, replay tool, and PDF are frozen together under the versioned tag `pf4-paper-v1.0.0`; their SHA-256 values are listed in `paper/RELEASE_MANIFEST.md`. The complete evidence replay is `python scripts/replay_paper.py` from the repository root.

All theorem-bearing sign decisions are exact. Numerical values produced by the repository scripts are diagnostic only; there is no separate experimental dataset.

## References

- [1] I. J. Schoenberg, *On Polya frequency functions. II. Variation-diminishing integral operators of the convolution type*, Acta Sci. Math. (Szeged) **12** (1950), 97–106.
- [2] S. Karlin, *Total Positivity, Vol. I*, Stanford University Press, Stanford, 1968.
- [3] A. Belton, D. Guillot, A. Khare, and M. Putinar, *Preservers of totally positive kernels and Polya frequency functions*, Math. Res. Rep. **3** (2022), 35–56, arXiv:2110.08206.
- [4] A. Khare, *Multiply positive functions, critical exponent phenomena, and the Jain–Karlin–Schoenberg kernel*, arXiv:2008.05121.
- [5] N. G. de Bruijn, *The roots of trigonometric integrals*, Duke Math. J. **17** (1950), 197–226, doi:10.1215/S0012-7094-50-01720-0.
- [6] D. K. Dimitrov and Y. Xu, *Wronskians of Fourier and Laplace transforms*, Constr. Approx. **48** (2018), 55–99, arXiv:1606.05011.
- [7] G. Csordas and R. S. Varga, *Moment inequalities and the Riemann Hypothesis*, Constr. Approx. **4** (1988), 175–198, doi:10.1007/BF02075457.
- [8] K. Grochenig, *Schoenberg’s theory of totally positive functions and the Riemann zeta function*, arXiv:2007.12889.
- [9] W. Michałowski, *On the Polya frequency order of the de Bruijn–Newman kernel: certified failure at order five and the Toeplitz threshold phenomenon*, arXiv:2602.20313.

## A Confluent and curvature algebra

The central-moment Hankel matrix is

$$\begin{pmatrix} 1 & 0 & m_2 & m_3 \\ 0 & m_2 & m_3 & m_4 \\ m_2 & m_3 & m_4 & m_5 \\ m_3 & m_4 & m_5 & m_6 \end{pmatrix}.$$

Taking the Schur complement of the upper-left entry reduces its determinant to

$$\det \begin{pmatrix} m_2 & m_3 & m_4 \\ m_3 & m_4 - m_2^2 & m_5 - m_2 m_3 \\ m_4 & m_5 - m_2 m_3 & m_6 - m_3^2 \end{pmatrix}.$$

Substitution of the moments from Section 4 and ordinary three-by-three expansion gives

$$\mathcal{C}_4 = 12c_2^6 + 24c_2^4c_4 - 24c_2^3c_3^2 + 2c_2^3c_6 - 12c_2^2c_3c_5 + 7c_2^2c_4^2 \quad (\text{A.1})$$

$$+ 12c_2c_3^2c_4 + c_2c_4c_6 - c_2c_5^2 - 9c_3^4 - c_3^2c_6 + 2c_3c_4c_5 - c_4^3. \quad (\text{A.2})$$

In the curvature coordinate, repeated application of  $Q d/dy$  to  $c_2 = -Q$  gives

$$\begin{aligned} c_2 &= -Q, \\ c_3 &= -QQ', \\ c_4 &= -Q\{(Q')^2 + QQ''\}, \\ c_5 &= -Q\{(Q')^3 + 4QQ'Q'' + Q^2Q'''\}, \\ c_6 &= -Q\{(Q')^4 + 11Q(Q')^2Q'' + 4Q^2(Q'')^2 + 7Q^2Q'Q''' + Q^3Q''''\}. \end{aligned}$$

Substitution in the preceding polynomial gives

$$\mathcal{C}_4 = -Q^6\mathcal{P},$$

where

$$\begin{aligned} \mathcal{P} &= QQ''Q'''' - Q(Q''')^2 - 2QQ'''' + Q'Q''Q''' - 2Q'Q''' \\ &\quad + 3(Q'')^3 - 15(Q'')^2 + 24Q'' - 12. \end{aligned}$$

Now  $\kappa = 2 - Q''$ , and direct differentiation gives

$$\{Q(\log \kappa)'\}' = \frac{Q'\kappa' + Q\kappa''}{\kappa} - \frac{Q(\kappa')^2}{\kappa^2},$$

so, after multiplication by  $\kappa^2$  and substitution of  $\kappa' = -Q'''$ ,  $\kappa'' = -Q''''$ ,

$$3(\kappa - 1) - \{Q(\log \kappa)'\}' = -\frac{\mathcal{P}}{(Q'' - 2)^2} = -\frac{\mathcal{P}}{\kappa^2}.$$

Therefore  $\mathcal{C}_4 = Q^6\kappa^2D$ , proving (8.2) by an exposed common numerator.

## B Exact proof of the certified input

This appendix proves Lemma 4.1. The positive-side theta series has  $j$ -th derivative

$$2\pi n^2 e^{5u/2 - n^2 x} P_j(n^2 x), \quad x = \pi e^{2u},$$

where the polynomials  $P_j$  satisfy the recurrence in Section 4. Machin's identity and alternating series give  $\pi > 157/50$ ; hence  $P_0(x) = 2x - 3 > 0$ .

### B.1 Cleared sign polynomials

In addition to  $a_j$ , define the normalized remaining modes

$$e_j(x) = \sum_{n \geq 3} \frac{n^2 e^{-(n^2 - 1)x} P_j(n^2 x)}{P_0(x)}, \quad b_j = a_j + e_j.$$

For an indeterminate raw jet  $(b_0, \dots, b_6)$ , put

$$N_q = b_1^2 - b_0 b_2,$$



$$N_{F_2} = -b_0^4 b_2 b_4 + b_0^4 b_3^2 + b_0^3 b_1^2 b_4 - 2b_0^3 b_1 b_2 b_3 + 2b_0^3 b_2^3 \\ - 3b_0^2 b_1^2 b_2^2 + 3b_0 b_1^4 b_2 - b_1^6,$$

and

$$N_{\mathcal{C}_4} = \det[b_{i+j}]_{i,j=0}^3.$$

Direct logarithmic differentiation and the central-moment determinant give

$$q = \frac{N_q}{b_0^2}, \quad F_2 = \frac{N_{F_2}}{b_0^6}, \quad \mathcal{C}_4 = \frac{N_{\mathcal{C}_4}}{b_0^4}. \quad (\text{B.1})$$

The thirteen-term cumulant expansion obtained from the last identity agrees with (A.1); this is an exact polynomial identity.

For orientation, retaining only the first mode gives

$$q_1 = \frac{4x(4x^2 - 12x + 15)}{(2x - 3)^2}, \\ (F_2)_1 = \frac{64x^3(64x^6 - 576x^5 + 2448x^4 - 6432x^3 + 10044x^2 - 8100x + 2295)}{(2x - 3)^6}, \\ (\mathcal{C}_4)_1 = \frac{49152x^6(16x^4 - 96x^3 + 360x^2 - 840x + 945)}{(2x - 3)^4}.$$

After  $x = 5 + z$ , every numerator has positive coefficients. The second mode is retained exactly because it supplies the larger margin near the origin.

## B.2 Exact two-mode margin

Alternating Taylor bounds for the exponential imply

$$0 < y < \frac{1}{12000} \quad (x \geq \pi),$$

and

$$\frac{1}{23000} < y < \frac{1}{12000} \quad \left( \frac{157}{50} \leq x \leq \frac{10}{3} \right).$$

We use the following elementary certificate lemma. If  $p(s, t) = \sum p_{ij} s^i t^j$  has bidegree  $(d, e)$ , its Bernstein coefficients on the unit square are

$$\beta_{k\ell} = \sum_{i \leq k, j \leq \ell} p_{ij} \frac{\binom{k}{i}}{\binom{d}{i}} \frac{\binom{\ell}{j}}{\binom{e}{j}}.$$

The power-to-Bernstein identity proves that  $\beta_{k\ell} > 0$  for every  $k, \ell$  implies  $p > 0$  on the square.

Apply this lemma after affine rational changes of variables. For  $q$ , one rectangle ends at  $x = 10/3$ , followed by a positive base polynomial minus exponentially decreasing corrections. For  $F_2$ , dependent bounds on the first rectangle are followed by a rectangle through  $x = 5$  and a positive half-strip expansion. For  $\mathcal{C}_4$ , the first rectangle is followed by two negative odd-power corrections whose ratios to the positive base decrease. The derivative of each ratio has a numerator with positive coefficients after the indicated shift. The 361 resulting rational coefficient inequalities give

$$q_{1,2} > 10, \quad (F_2)_{1,2} > 1000, \quad (\mathcal{C}_4)_{1,2} > 50,000,000. \quad (\text{B.2})$$

This is a finite algebraic proof on fixed domains.

For reference, the complete finite certificate is summarized below. Write  $\mathcal{A} = (2x-3)+4(8x-3)y$ , which is positive on every listed domain. Rows  $Q_0, F_0, F_1, C_0$  are Bernstein certificates after the indicated affine change to the unit square; the remaining rows use coefficient positivity after a half-line shift. In the two “ratio” rows, the listed fraction is the sum of the adverse coefficient ratios and is strictly less than one.

| ID         | Variables and domain                                   | Degree  | Positive denominator | Exact lower datum                          |
|------------|--------------------------------------------------------|---------|----------------------|--------------------------------------------|
| $Q_0$      | $157/50 \leq x \leq 10/3, 0 \leq y \leq 1/12000$       | (4, 2)  | $\mathcal{A}^2$      | 6613625187767/70312500000                  |
| $Q_\infty$ | $x = 10/3 + z, 0 \leq y \leq 1/12000$                  | (4, 2)  | $\mathcal{A}^2$      | ratio 47333/505500                         |
| $F_0$      | $157/50 \leq x \leq 10/3, 1/23000 \leq y \leq 1/12000$ | (12, 6) | $\mathcal{A}^6$      | minimum > 1610047                          |
| $F_1$      | $10/3 \leq x \leq 5, 0 \leq y \leq 1/22000$            | (12, 6) | $\mathcal{A}^6$      | 29124973000/19683                          |
| $F_\infty$ | $x = 5 + z, y = v/3000000$                             | (12, 6) | $\mathcal{A}^6$      | 432/19073486328125                         |
| $C_0$      | $157/50 \leq x \leq 10/3, 0 \leq y \leq 1/12000$       | (14, 4) | $\mathcal{A}^4$      | minimum > 9546672947                       |
| $C_\infty$ | $x = 10/3 + z, 0 \leq y \leq 1/12000$                  | (14, 4) | $\mathcal{A}^4$      | ratio 65989337396066791/771657205500000000 |

These seven checks comprise 361 exact rational inequalities. The derivative envelopes used below contribute 140 further Bernstein coefficients, of bidegrees  $(j+1, 1)$ ,  $0 \leq j \leq 6$ ; their smallest coefficient is 44861/31250. Thus the compact domains and the differentiated tail majorants belong to the same exact certificate.

### B.3 All later modes

For  $t \geq 27$ , shifting  $t = 27 + z$  gives positive coefficients in  $(-1)^j P_j(t)$  and in  $2^{j+1}t^{j+1} - (-1)^j P_j(t)$ . Therefore

$$0 < (-1)^j P_j(t) \leq 2^{j+1}t^{j+1}, \quad 0 \leq j \leq 6. \quad (\text{B.3})$$

The derivative numerator of the normalized  $n = 3$  term is another positive shifted polynomial, so its absolute value decreases for  $x \geq 3$ . For  $n \geq 4$ , successive majorants have ratio at most

$$\left(\frac{5}{4}\right)^{16} e^{-27} < 10^{-9}.$$

Rational Taylor lower bounds for  $e^{24}, e^{27}, e^{45}$  then yield, on  $\pi \leq x \leq 5$ ,

$$(|e_0|, \dots, |e_6|) < (7 \cdot 10^{-9}, 4 \cdot 10^{-7}, 2 \cdot 10^{-4}, 7 \cdot 10^{-4}, 3 \cdot 10^{-2}, 1.1, 41). \quad (\text{B.4})$$

The same Bernstein lemma gives

$$(|a_0|, \dots, |a_6|) < (2, 5, 20, 200, 2000, 60000, 1000000).$$

Expanding the three polynomials in (B.1) and using absolute coefficients gives

$$|\Delta N_q| < 0.000405, \quad |\Delta N_{F_2}| < 37.959, \quad |\Delta N_{C_4}| < 31,549,532. \quad (\text{B.5})$$

For  $x \geq 5$ , (B.3) gives

$$|a_j| \leq 2^{j+2}x^j, \quad |e_j| \leq 2^{j+2}3^{2j+4}x^j e^{-8x}.$$

The raw weights of  $N_q, N_{F_2}, N_{C_4}$  are respectively 2, 6, 12. Thus every perturbation majorant is a positive multiple of  $x^w e^{-8x}$ , or a faster-decaying term, with  $w \leq 12$ ; it decreases for  $x \geq 5$ . A single rational Taylor bound at  $x = 5$  gives

$$|\Delta N_q| < 6.49 \cdot 10^{-11}, \quad |\Delta N_{F_2}| < 0.03, \quad |\Delta N_{C_4}| < 418,287. \quad (\text{B.6})$$

To compare like quantities, note that

$$a_0 = 1 + 4y \frac{P_0(4x)}{P_0(x)} > 1,$$

and, by homogeneity,

$$N_{q,1,2} = q_{1,2}a_0^2, \quad N_{F_2,1,2} = (F_2)_{1,2}a_0^6, \quad N_{C_4,1,2} = (C_4)_{1,2}a_0^4.$$

Consequently (B.2) supplies the same respective lower margins 10, 1000, and 50,000,000 for the cleared numerators. Both perturbation bounds (B.5) and (B.6) are strictly smaller. Finally  $b_0 = a_0 + e_0 > 0$ , so (B.1) proves all three signs for  $u \geq 0$ . Evenness proves Lemma 4.1.

## C Endpoint algebra for the transport cancellation

This appendix records the algebra suppressed by the phrase “simplifying the endpoint expression.” Put

$$z = p + L, \quad w = z + R, \quad a = \frac{Q(z) - Q(p)}{L}, \quad b = \frac{Q(w) - Q(z)}{R}.$$

Direct integration of  $2 - Q''$  gives

$$\delta = \frac{L + Q'(p) - a}{L}, \quad \Lambda = L + R + a - b. \quad (\text{C.1})$$

Define

$$J_y = y^3 - 3yQ(y) + Q(y)Q'(y), \\ H_p = 3p^2 - 3Q(p) - 3pQ'(p) + Q'(p)^2 + Q(p)Q''(p),$$

and, for endpoint data,

$$U(c, d) = Q(c) + Q(d) - \frac{Q(d)Q'(d) - Q(c)Q'(c)}{d - c} + \frac{(Q(d) - Q(c))^2}{(d - c)^2}.$$

With  $I_L = J_z - J_p$  and  $I_R = J_w - J_z$ , the identity needed in Lemma 9.1 is exactly

$$\begin{aligned} & \frac{-(J_z - J_p)/L + (J_w - J_z)/R}{\Lambda} + \frac{LH_p - (J_z - J_p)}{L^2\delta} \\ &= \Lambda + Q'(p) - a + \frac{U(p, z) - U(z, w)}{\Lambda} + \frac{U(p, z) - Q(p)(2 - Q''(p))}{L\delta}. \end{aligned}$$

To check it, substitute (C.1), replace  $z$  by  $p + L$  and  $w$  by  $p + L + R$ , and multiply by the common denominator  $L^2R\delta\Lambda$ . The terms containing  $Q'(w)$ , then  $Q'(z)$ , then  $Q''(p)$  cancel pairwise; the remaining endpoint values and position terms cancel after inserting  $a = (Q(z) - Q(p))/L$  and  $b = (Q(w) - Q(z))/R$ . No differential equation for  $Q$  is used, so the identity holds for arbitrary smooth  $Q$ .

## D Certificate inventory

The finite and symbolic parts of the proof are recorded in the following plain-text source files. Each record states the exact claim, its inputs, and the corresponding verifier.

| Claim                                               | Proof record                                                 |
|-----------------------------------------------------|--------------------------------------------------------------|
| Global $q, F_2, \mathcal{C}_4$ , order three        | <code>sources/riemann-signs-analytic-certificate.md</code>   |
| Quotient and three-point equivalence                | <code>sources/pf4-wronskian-reduction-certificate.md</code>  |
| Transport identity and completion                   | <code>sources/pf4-transport-kernel-certificate.md</code>     |
| Origin order-five obstruction                       | <code>sources/pf5-origin-certificate.md</code>               |
| Continuous strict-PF <sub>4</sub> Fourier separator | <code>sources/continuous-pf4-separator-certificate.md</code> |

The maintained source map `paper/PAPER_MAP.md` identifies the manuscript section in which each record is used. The single replay command in Section 13 checks the five corresponding verifier outputs.

## E Exact separator calculation

This appendix records the algebraic boundary behind (12.7). Starting from the cumulants  $\ell_2, \dots, \ell_6$ , the verifier reconstructs the central moments

$$\begin{aligned}
m_0 &= 1, & m_1 &= 0, & m_2 &= \ell_2, & m_3 &= \ell_3, \\
m_4 &= \ell_4 + 3\ell_2^2, & m_5 &= \ell_5 + 10\ell_3\ell_2, \\
m_6 &= \ell_6 + 15\ell_4\ell_2 + 10\ell_3^2 + 15\ell_2^3,
\end{aligned}$$

and then forms  $\det[m_{i+j}]_{i,j=0}^3$  directly. This independently regenerates the thirteen-term  $\mathcal{C}_4$  polynomial; it does not import the expansion used for the Riemann kernel.

The recurrence (12.5) is then applied over  $\mathbb{Q}[t]$ . Substitution of (12.6) shows that each of the thirteen determinant monomials has common denominator  $P^{12}$ . After collecting, the numerator is divisible by  $P^8$  in  $\mathbb{Q}[t]$ . Cancel this factor and clear the remaining powers of two, writing

$$\mathcal{C}_4[f_*](x) = \frac{\tilde{H}(t)}{2^{54}P(t)^4}.$$

Exact rational arithmetic verifies

$$\deg \tilde{H} = 24, \quad [t^j]\tilde{H} = [t^{24-j}]\tilde{H} > 0 \quad (0 \leq j \leq 24),$$

with smallest integer coefficient

$$\min_{0 \leq j \leq 24} [t^j]\tilde{H} = 3221225472.$$

Equivalently, the smallest coefficient before multiplication by  $2^{54}$  is  $3/2^{24}$ . Write

$$\tilde{H}(t) = \sum_{j=0}^{24} a_j t^j, \quad a_j = a_{24-j}.$$

The 13 independent integer coefficients are:

| $j$ | $a_j$           | $j$ | $a_j$            |
|-----|-----------------|-----|------------------|
| 0   | 3221225472      | 7   | 357704506872240  |
| 1   | 61435985920     | 8   | 701640591264080  |
| 2   | 575070935296    | 9   | 1172880923200640 |
| 3   | 3512452441920   | 10  | 1684288128586089 |
| 4   | 15702254784944  | 11  | 2088645133531440 |
| 5   | 54633824917920  | 12  | 2243243092164284 |
| 6   | 153607464071178 |     |                  |

This table is generated by `scripts/generate_separator_coefficients.py`. Its machine-readable companion `generated/separator-coefficients.json` has SHA-256 prefix `d0f91259919c7f23`; the complete hash is recorded in the release manifest. Coefficient positivity proves the sign on the full domain  $t > 0$ . Exact spot reconstructions at  $t = 1/3, 1, 4$  independently compare the central determinant with  $\tilde{H}/(2^{54}P^4)$ .

For the Fourier step, the identities

$$w + w^{-1} = u, \quad w^2 + w^{-2} = u^2 - 2, \quad w^3 + w^{-3} = u^3 - 3u$$

give (12.9) directly. Put

$$r = e^{1/64}, \quad x = r^2 = e^{1/32}.$$

Exact expansion gives

$$\text{disc}(R_c) = 4r^{10}G(x),$$

where

$$\begin{aligned} G(x) = & 432x^{13} - 4968x^9 + 900x^8 + 16200x^6 + 19044x^5 \\ & - 72600x^4 + 20000x^3 + 62100x^2 - 54334x + 13225. \end{aligned}$$

Writing  $y = x - 1$  gives

$$\begin{aligned} G(1+y) = & -1 - 10y - 12y^2 + 680y^3 + 11532y^4 + 96660y^5 \\ & + 365400y^6 + 569664y^7 + 512172y^8 + 303912y^9 \\ & + 123552y^{10} + 33696y^{11} + 5616y^{12} + 432y^{13}. \end{aligned}$$

Moreover  $0 < y < 1/30$ . Indeed,

$$e < 1 + 1 + \frac{1}{2} + \frac{1}{6} + \frac{1}{18} = \frac{49}{18} < \frac{11}{4},$$

where  $1/18 = (1/24) \sum_{j \geq 0} 4^{-j}$  bounds the tail beginning with  $1/4!$ . The binomial theorem gives

$$\frac{11}{4} < 1 + \frac{32}{30} + \frac{\binom{32}{2}}{30^2} + \frac{\binom{32}{3}}{30^3} = \frac{1891}{675} < \left(\frac{31}{30}\right)^{32}.$$

The sum of the positive terms at  $y = 1/30$  is exactly

$$\sum_{j=3}^{13} \frac{[y^j]G(1+y)}{30^j} = \frac{1621193195302591}{36905625000000000} < 1.$$

The two omitted terms  $-10y - 12y^2$  are negative. Consequently

$$G(1+y) < -1 + \sum_{j=3}^{13} \frac{[y^j]G(1+y)}{30^j} < 0,$$

and hence  $\text{disc}(R_c) < 0$  without numerical evaluation.